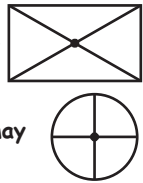


Centre of Mass

• Avg. position of all the parts of the system, weighted according to their mass

• For homogeneous objects, centre of mass lies at their geometric centre

• Centre of mass may or may not lie inside the object



Centre of Mass For System of n Particles

$$\vec{r}_{cm} = \frac{\sum m_i \vec{r}_i}{\sum m_i}$$

For n particles

General Equation

$$\vec{r}_{cm} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2 + m_3 \vec{r}_3 + \dots + m_n \vec{r}_n}{m_1 + m_2 + m_3 + \dots + m_n}$$

In terms of Cartesian co-ordinates

$$x_{cm} = \frac{m_1 x_1 + m_2 x_2 + m_3 x_3 + \dots + m_n x_n}{m_1 + m_2 + m_3 + \dots + m_n}$$

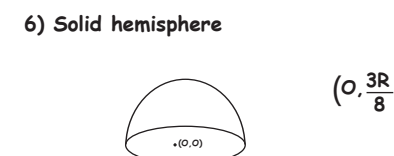
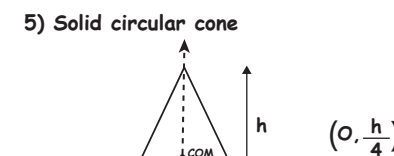
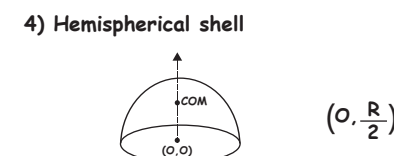
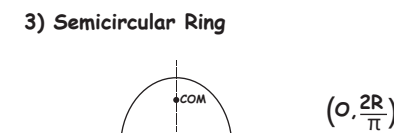
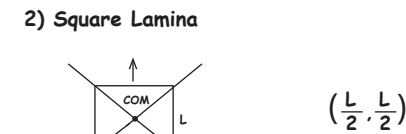
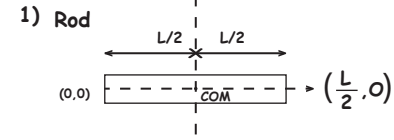
$$y_{cm} = \frac{m_1 y_1 + m_2 y_2 + m_3 y_3 + \dots + m_n y_n}{m_1 + m_2 + m_3 + \dots + m_n}$$

$$z_{cm} = \frac{m_1 z_1 + m_2 z_2 + m_3 z_3 + \dots + m_n z_n}{m_1 + m_2 + m_3 + \dots + m_n}$$

Centre of mass for various shapes

Uniformly distributed mass

centre of mass



Motion of centre of mass

velocity of centre of mass

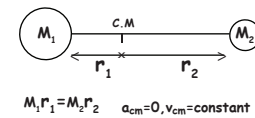
$$\vec{V}_{cm} = \frac{M_1 \vec{V}_1 + M_2 \vec{V}_2 + M_3 \vec{V}_3 + \dots}{M_1 + M_2 + M_3 + \dots}$$

Acceleration of centre of mass

$$\vec{a}_{cm} = \frac{M_1 \vec{a}_1 + M_2 \vec{a}_2 + M_3 \vec{a}_3 + \dots}{M_1 + M_2 + M_3 + \dots}$$

Isolated System

• No net external force acting on the system
• Bodies within the system can have mutual force between them



$$M_1 r_1 = M_2 r_2 \quad a_{cm} = 0, v_{cm} = \text{constant}$$

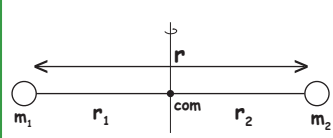
Moment of Inertia (for a point object)

$I = mr^2$
m = Mass of body
r = Perpendicular distance of the body from the axis of rotation

Moment of Inertia
Tensor Quantity $I = mr^2$ Rotational analogous of mass

Two Point Masses

$$I_{com} = m_1 r_1^2 + m_2 r_2^2$$



$$r_1 = \frac{m_2 r}{m_1 + m_2}, \quad r_2 = \frac{m_1 r}{m_1 + m_2}$$

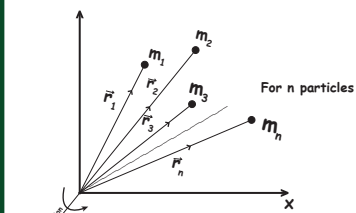
$$I_{com} = m_{red} r^2, \quad m_{red} = \frac{m_1 m_2}{m_1 + m_2}$$

Factors Affecting Moment of Inertia

Mass of the body, Axis of rotation, Mass distribution

Moment of Inertia

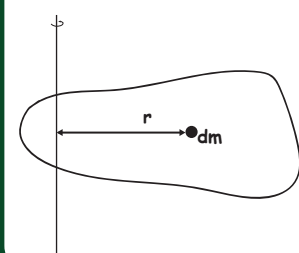
i) for discrete system of particles



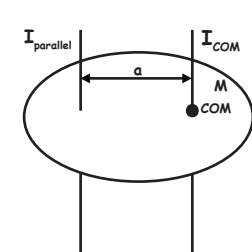
$$I = \frac{m_1 r_1^2 + m_2 r_2^2 + m_3 r_3^2 + \dots + m_n r_n^2}{m_1 + m_2 + m_3 + \dots + m_n}$$

ii) for continuous body

$$I = \int dI = \int r^2 dm$$



Parallel Axis Theorem



$$I_{parallel} = I_{com} + Ma^2$$

Conditions:-

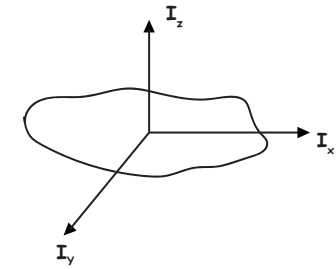
- 1) the two axes must be parallel to each other
- 2) One of the axis must pass through centre of mass

Perpendicular axis theorem

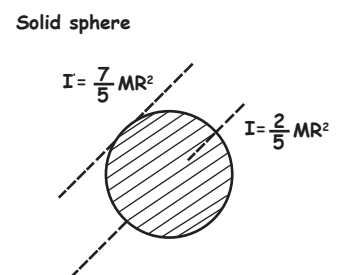
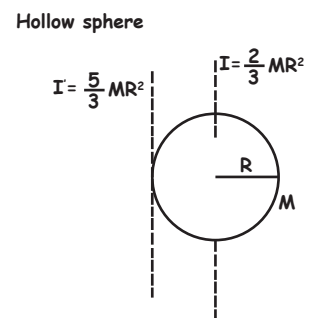
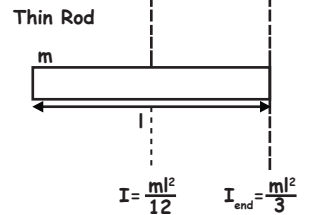
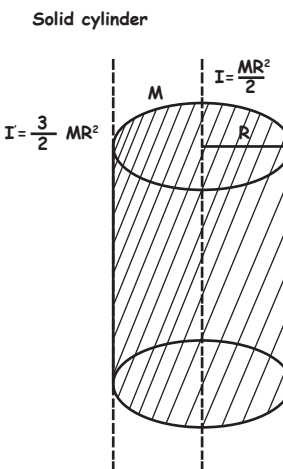
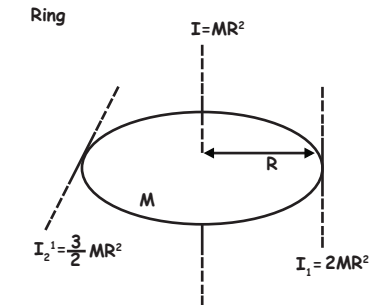
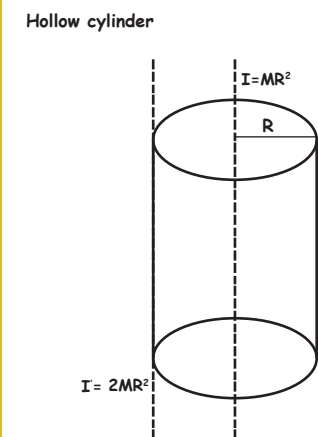
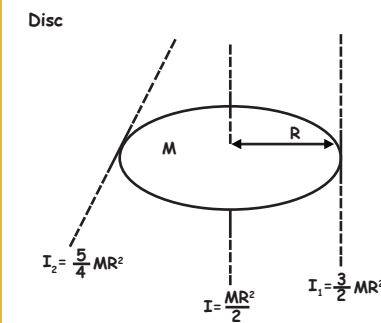
$$I_z = I_x + I_y \quad (\text{Only valid for laminar bodies})$$

Note:-

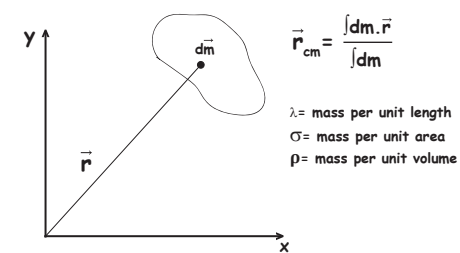
X and Y axis must lie in the plane of body
Z-axis must be $\perp$ to the plane of the body
Axes need not pass through center of mass



Moment of Inertia For Various Objects



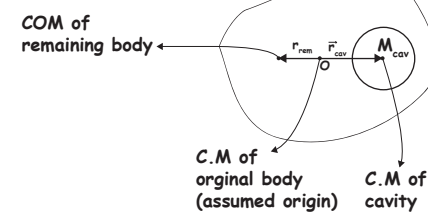
Centre Of Mass For Continuous Body



- 1) Mass distributed over length $\Rightarrow dm = \lambda dl$
- 2) Mass distributed over area $\Rightarrow dm = \sigma dA$
- 3) Mass distributed over volume $\Rightarrow dm = \rho dV$

Cavity in object

If some mass is removed from a body, COM will shift towards the side with more mass



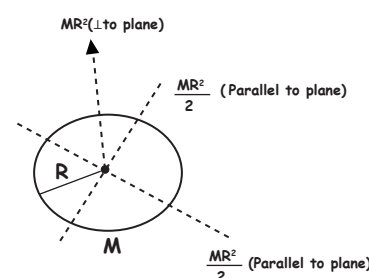
Assuming COM of original body is at the origin

$$\vec{r}_{rem} = \frac{-M_{cav} \times \vec{r}_{cav}}{M_{rem}}$$

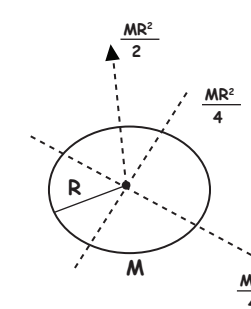
ROTATIONAL MOTION 01

Moment of Inertia along the centre of mass and perpendicular to the plane surface

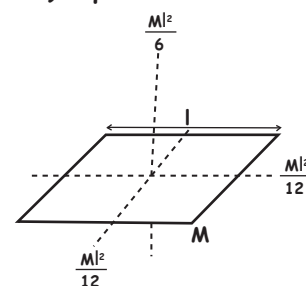
1) Ring



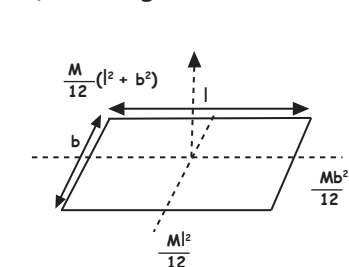
2) Disc



3) Square sheet

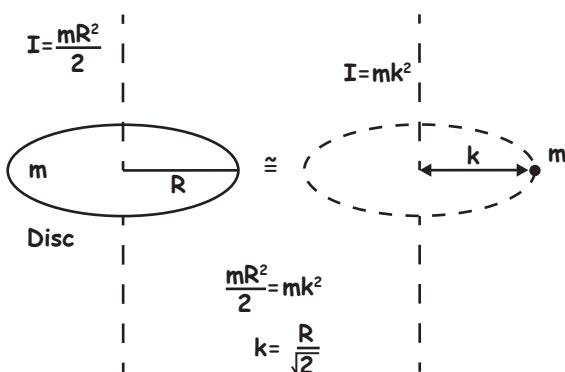


4) Rectangular sheet



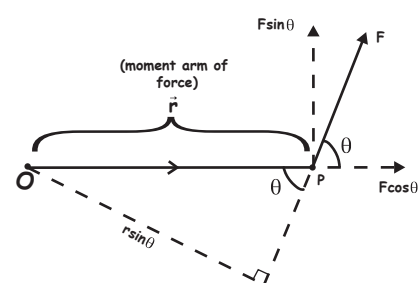
Radius of Gyration

Definition: The distance of a point mass from the axis whose mass is equal to the mass of whole body and whose moment of inertia is equal to moment of inertia of the body about that axis



k is the radius of gyration

Torque



$$\text{Torque } \tau_o = r F \sin \theta$$

$$= F \sin \theta \times r = F_{\perp} r \quad (1)$$

$$= F \times r \sin \theta = F_{\perp} r \quad (2)$$

$$\vec{\tau}_o = \vec{r} \times \vec{F} \text{ (Vector form)}$$

If force is radial i.e. $\theta = 0^\circ$ or 180°

Torque $\tau = 0$

If force is tangential and $\perp$ to radius vector i.e. $\theta = 90^\circ$

$$\text{Torque } \tau = \tau_{\max} = rF$$

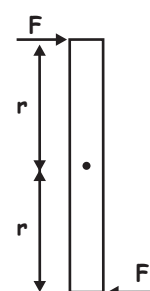
ROTATIONAL MOTION 02

Equilibrium

For translational equilibrium

$$F_{\text{net}} = 0$$

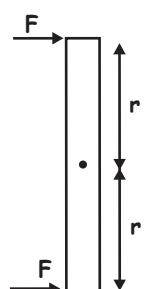
τ_{net} may or maynot be zero



Rotational Equilibrium

$$\tau_{\text{net}} = 0$$

F_{net} may or maynot be zero



Static Equilibrium

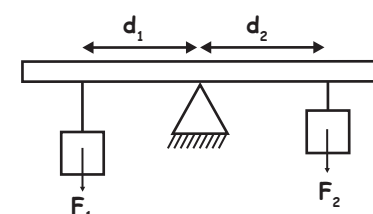
Combination of both translational and rotational equilibrium

$F_{\text{net}} = 0 \Rightarrow$ Forces are balanced

$\tau_{\text{net}} = 0 \Rightarrow \tau_{\text{clockwise}} = \tau_{\text{anticlockwise}}$

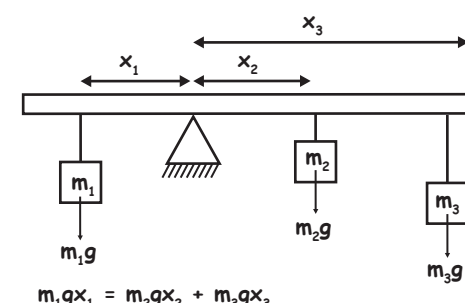
Principle of moments

When a body is in rotational equilibrium sum of clock wise moments about any point is equal to sum of anticlockwise moments about that point

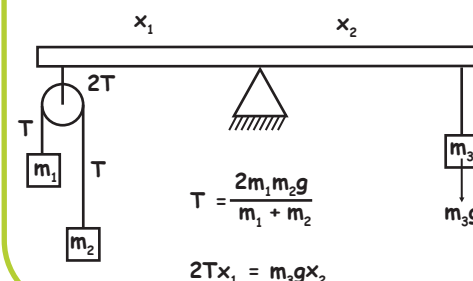


$$F_1 \times d_1 = F_2 \times d_2$$

Load $\times$ load arm = Effort $\times$ effort arm



$$m_1 g x_1 = m_2 g x_2 + m_3 g x_3$$



$$T = \frac{2m_1 m_2 g}{m_1 + m_2}$$

$$2T x_1 = m_3 g x_2$$

Angular acceleration

$$\tau = I\alpha$$

τ - torque

I moment of inertia

α angular acceleration

Initial angular acceleration when a rod is released

Initial angular acceleration when a body is released from an angle θ

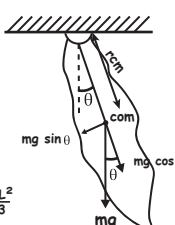
$$\tau = I\alpha$$

$$(mg \sin \theta) r_{\text{cm}} = I\alpha$$

$$\alpha = \frac{(mg \sin \theta) r_{\text{cm}}}{I}$$

$$\text{For rod } r_{\text{cm}} = \frac{L}{2}$$

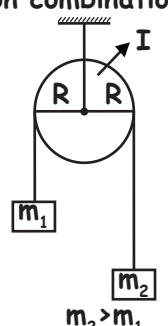
$$I = \frac{ML^2}{12}$$



Translation - rotation combination

$$\alpha = \frac{\tau_{\text{applied}} - \tau_{\text{opposition}}}{\text{Total } I}$$

$$\alpha = \frac{m_2 R g - m_1 R g}{m_1 R^2 + m_2 R^2 + I}$$



$$m_2 > m_1$$

Angular momentum & its conservation

Angular momentum of a point mass:-

Angular momentum about origin

$$\vec{L} = \vec{r} \times \vec{p}$$

$$= m(\vec{r} \times \vec{v})$$

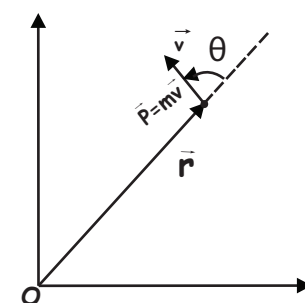
$$\vec{p} = m\vec{v}$$

1) When $\theta = 0^\circ$ or 180°

$$L_o = mvr \sin 180^\circ = 0$$

$$\text{OR } mvr \sin 0^\circ = 0$$

Angular momentum is minimum



1) When $\theta = 90^\circ$

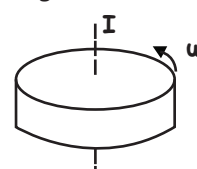
$$|\vec{L}| = r p \sin \theta$$

$$= r p \sin 90^\circ$$

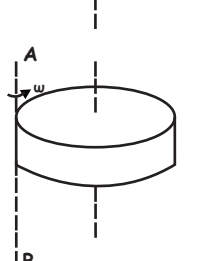
$$= r p$$

$$= L_{\max}$$

Spin angular momentum



$$L_{\text{axis}} = I\omega$$



$$L_{AB} = I_{AB}\omega$$

Conservation of Angular momentum

If there is no external torque, angular momentum is conserved

$$\tau = \frac{dL}{dt}$$

$$\text{If } \tau = 0 \Rightarrow \frac{dL}{dt} = 0$$

$$L = \text{constant}$$

$$I\omega = \text{constant}$$

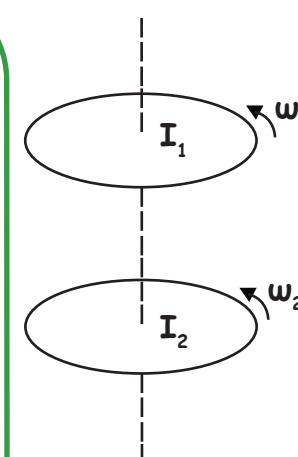
$$I_1 \omega_1 = I_2 \omega_2$$

If moment of inertia increases angular velocity decreases and if moment of inertia decreases angular velocity increases

Moment of inertia when two discs are joined

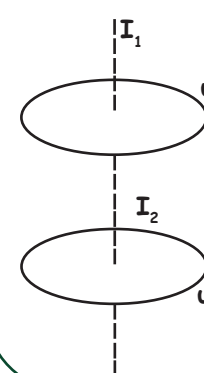
Discs initially rotating in same direction:-

$$\omega_f = \frac{I_1 \omega_1 + I_2 \omega_2}{I_1 + I_2}$$



Discs initially rotating in opposite direction:-

$$\omega_f = \frac{I_1 \omega_1 - I_2 \omega_2}{I_1 + I_2}$$



Work, Energy & Power in rotation

1) Work done by a torque,

$$W = \tau \theta \text{ (if torque is uniform)}$$

$$= \int \tau d\theta \text{ (if torque is non uniform)}$$

2) K.E for are rotating body $= \frac{1}{2} I \omega^2$

$$= \frac{1}{2} I \omega^2$$

$$= \frac{1}{2} I \omega^2$$

$$= \frac{1}{2} I \omega^2$$

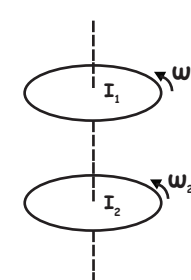
3) Work-Energy theorem

$$\Sigma W = \Delta K = \frac{1}{2} I (\omega_2^2 - \omega_1^2)$$

Energy loss when 2 discs are joined:-

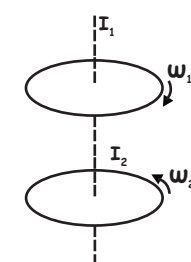
1) Same direction:-

$$E_{\text{lost}} = \Delta K.E = \frac{I_1 I_2}{2(I_1 + I_2)} (\omega_1 - \omega_2)^2$$



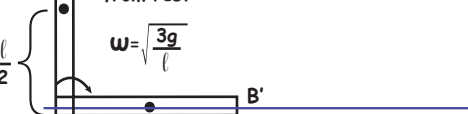
2) Opposite direction:-

$$E_{\text{lost}} = \Delta K.E = \frac{I_1 I_2}{2(I_1 + I_2)} (\omega_1 + \omega_2)^2$$



Mechanical energy conservation

Angular velocity with which the rod hits the ground without slipping, released from rest



Rolling Motion
Translatory + Rotatory = Rolling

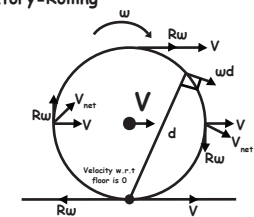
Velocity in rolling

Condition for rolling without slipping:- $V = R\omega$

Velocity of any point on rolling object, V_p

$$= \omega d = \frac{v d}{R}$$

d is the distance from point of contact



Energy in rolling motion

1) Translatory Motion

$$K_{\text{trans}} = \frac{1}{2} m v^2$$

2) Spinning motion/rotational motion

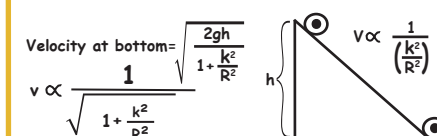
$$K_{\text{rot}} = \frac{1}{2} I \omega^2 = \frac{1}{2} m k^2 \frac{V^2}{R^2} = \frac{1}{2} m v^2 \times \left(\frac{k^2}{R^2} \right)$$

3) Rolling motion

$$K_{\text{trans}} + K_{\text{rot}} = \frac{1}{2} m v^2 + \frac{1}{2} m v^2 \times \frac{k^2}{R^2} = \frac{1}{2} m v^2 \left(1 + \frac{k^2}{R^2} \right)$$

$$\frac{K_{\text{total}}}{K_{\text{trans}}} = \left(1 + \frac{k^2}{R^2} \right)$$

Motion on an inclined plane



$$V \propto \frac{1}{\sqrt{1 + \frac{k^2}{R^2}}}$$

$$\frac{k^2}{R^2} \uparrow \Rightarrow V \downarrow \Rightarrow \text{Time } \uparrow$$

Velocity: solid sphere > Disc > Hollow sphere > Ring

Time to reach bottom: Ring > Hollow sphere > Disc > solid sphere

Value of velocity:-

$$1) \text{ Ring/Hollow cylinder} = \sqrt{\frac{gh}{2}}$$

$$2) \text{ Disc/Solid cylinder} = \sqrt{\frac{4}{3} gh}$$

$$3) \text{ Hollow sphere} = \sqrt{\frac{6}{5} gh}$$

$$4) \text{ Solid sphere} = \sqrt{\frac{10}{7} gh}$$

$$\text{Acceleration}$$

$$a = \frac{g \sin \theta}{1 + \frac{k^2}{R^2}} \quad a \propto \frac{1}{1 + \frac{k^2}{R^2}}$$

$$\text{Time of descend:-}$$

$$t = \frac{1}{\sin \theta} \sqrt{\frac{2h}{g} \left(1 + \frac{k^2}{R^2} \right)}$$

$$t \propto \sqrt{1 + \frac{k^2}{R^2}}$$

$$\text{Ring} > \text{Hollow sphere} > \text{Disc} > \text{solid sphere}$$